

Working Notes: Fruitfulness, Compression, and the Shape of Mathematics

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“Civilization advances by extending the number of important operations which we can perform
without thinking about them.”
— A. N. Whitehead, 1911

1 Where we are

We asked an open question, *what is the fruitfulness of a mathematical idea?*, and gave it a precise answer.

The definition. A named concept (a definition or a lemma) is a *compression*: its body is written once and cited afterwards.

What is proved in Lean. No sorry anywhere.

- Naming a body of size b that is used k times saves exactly $(k - 1)(b - 1) - 1$ symbols.
- Unfolding every citation at most doubles the size per result.
- That doubling is attained, and a Fibonacci library grows by the golden ratio per result.

What we measured. We exported all 771,129 constants of Mathlib, with 20.4M citation edges, and analysed the 315,085 user-facing ones.

- **Naive unfolding explodes.** Unfolded size grows about $1.5\times$ per dependency level ($R^2 = 0.98$). At the deepest level it reaches roughly $10^{65.6}$ symbols.
- **Reuse rediscovers the textbook.** Ranked by reuse, the top concepts are `Set`, `Category`, `Functor`, `TopologicalSpace`, `Module`, and so on.
- **Two axes of human taste, each computable.**
 - Concepts that humans call important are predicted by *reuse*: AUC 0.87, or 0.86 when compared only within the same file.
 - Theorems that humans call famous are predicted by *depth*: AUC 0.78, or 0.66 within the same file.

These notes record what I now think these results *mean*. Most of it is questions.

2 A correction I owe first

We said “fully unfolded, Mathlib does not fit in the universe.” That is true of *naive* inlining. But U is only an *upper* bound on the size of a lemma-free proof. A clever re-prover can restructure a proof instead of copying lemma bodies.

For Frege systems this is a theorem: tree-like Frege polynomially simulates dag-like Frege (Krajíček). There, reusing lemmas buys at most a *polynomial* saving, even though naive unfolding is exponential. Our exponential theorems are correct, but they are about inlining, not about the minimal lemma-free proof. The honest claim is:

- *lemma names* make mathematics exponentially shorter than *copying*;
- whether *some* lemma-free proof would be short is a question of **proof complexity**, and it is largely open.

3 The deepest form of our question is a famous open problem

There are two kinds of naming, and they are not the same.

1. **Lemmas** name a *proved statement*: reuse a result. In proof complexity this is dag-like versus tree-like proofs.
 - For resolution, the gap is exponential (Ben-Sasson, Impagliazzo and Wigderson).
 - For Frege, the gap is polynomial (Krajíček).
2. **Definitions** introduce a *new symbol* that abbreviates a formula. That is Tseitin’s extension rule, and Extended Frege is Frege plus definitions.
 - Whether Extended Frege is super-polynomially stronger than Frege is one of the central open problems of proof complexity.

Question (Q1, sharpened). *Are new concepts ever exponentially necessary, beyond reusing lemmas? Or can every proof that uses definitions be rewritten, at polynomial cost, using only old vocabulary?*

This is exactly the question “are human concepts forced or merely convenient?”, in formal dress.

Going up a type level, the answer is spectacular. Gödel (1936) showed that moving to higher-order logic shortens some proofs by more than any computable function. Boolos’s “curious inference” (1987) is a concrete case: a valid first-order argument whose shortest first-order proof is unimaginably long, physically impossible to write down, while a short second-order proof exists. So at the level of *types of concepts*, concepts are provably forced. At the propositional level, nobody knows.

Question (Q1’). *Mathlib is a vast empirical sample of definitions in use. Can its data say anything about where definitions give super-polynomial savings? That would give experimental evidence on a problem that has resisted proof.*

4 Value has two axes: depth and leverage

Our human-label results split cleanly.

- **Famous theorems are deep**, in the sense of $\log U - \log s$. They certify a large amount of hidden work behind a short statement.
- **Fundamental concepts have leverage**. They are small and reused enormously.

Both axes already exist in the theory of information, independently of mathematics.

- **Bennett’s logical depth** (1988). An object is valuable when it is the result of a long computation that cannot be shortcut: a genome, a solved theorem. Bennett proposed depth, not information content, as the measure of value. Our finding that fame tracks unfolded depth is, as far as I know, the first large-scale test of Bennett’s thesis on human judgments of mathematical value.

- **Schmidhuber’s compression progress** (2009). An observer finds something interesting when it improves the observer’s compressor; the first derivative of compressibility. Our $(k-1)(b-1)-1$ is exactly the compression progress a single new concept produces. Reuse predicting human concept taste (0.87) is evidence for Schmidhuber’s thesis in mathematics.

Question (Q2). *Are depth and leverage the only two axes of mathematical value? What explains the residual, the roughly 15–20% of human judgments neither measure predicts? Is it surprise, beauty, connection between distant fields, or just historical accident? Can a third measure, such as bridging (citations spanning distant parts of the library), absorb it?*

This bears directly on the debate that started this. “Nothing to optimize” is false in a strong sense. Theories of the *reward* for open-ended inquiry exist (Bennett, Schmidhuber), and here they predict human taste.

5 A glue law across nature

To see meaning in Mathlib we had to remove a layer of *typeclass plumbing*:

- instance lemmas that are used thousands of times;
- each one carries almost no content;
- no human would ever call one important.

The same layer appears everywhere reuse appears.

- **Language.** The most frequent words are *function words* (*the, of*). They are pure glue, and corpus linguists filter them out as “stop words”.
- **Metabolism.** The hubs of metabolic networks are *currency metabolites* (ATP, NADH, water). Network biologists must remove them before pathways become visible (Jeong et al. 2000; Ma and Zeng 2003).
- **Software.** The most depended-upon packages are tiny utilities. The `left-pad` incident is the famous case.
- **Mathlib.** Coercions, instance projections, `mul_one`.
There are more parallels.
- **Heavy-tailed reuse** appears in all of these. Mathlib has $P(k \geq x) \propto x^{-0.8}$, with 165 constants used more than 10,000 times. Compare Zipf’s law for words, protein-domain family sizes (Koonin et al.), and citation counts (Price’s cumulative advantage).
- **The law of abbreviation** holds as well. In natural language, frequent words are shorter (Zipf). We just checked Mathlib, and it holds there too.
 - The mean name length of definitions falls monotonically with use: from 13.4 characters (never used) to 7.1 (used 10^4 – 10^5 times) to 3 (`Set`, `coe`).
 - Spearman $\rho(\text{uses}, \text{name length}) = -0.21$.
 - Spearman $\rho(\text{uses}, \text{body size}) = -0.25$.
 Nobody designed this. Mathlib’s naming convention derives names from statements, so the law emerges from which concepts get reused.

Question (Q3: the glue theorem). *Is there a single theorem behind all of these? Consider any system that grows by reuse under a cost on description length. Must it develop (i) heavy-tailed reuse, (ii) a high-frequency, low-content glue layer, and (iii) abbreviation of the frequent? Preferential attachment explains (i), and Zipf’s least-effort argument gestures at (iii). A unified compression-theoretic derivation of all three would connect linguistics, metabolism, software and mathematics.*

Why Mathlib matters here. It is the *model organism* of reuse systems.

- In biology, edges are inferred and noisy. In language, “concept” is fuzzy.
- In Mathlib, every edge is *exact* and every unit is *checked*.

That makes laws testable here first, with unusual precision.

6 Hierarchy, decoupling, and why the world is compressible

Simon’s watchmakers. Herbert Simon’s *Architecture of Complexity* (1962) tells of two watchmakers.

- One assembles stable sub-assemblies. The other assembles parts one by one.
- Interrupted often, only the first finishes. The hierarchical builder has an exponential advantage.
- Simon argued this is why nature is hierarchical: cells, organs, organisms, organizations.

Our Lean theorems are the *size* counterpart of his *time* argument.

Proof irrelevance is scale separation. In Lean, any two proofs of the same proposition are equal. A later result can depend only on the *statement* of a lemma, never on its proof. That is precisely the physicist’s *decoupling of scales*.

- Chemistry needs the periodic table, not quantum chromodynamics.
- Renormalization explains why the microscopic details wash out.
- The quantity $\log U - \log s$ measures how much “microstate” a statement “macrostate” hides: an entropy of forgetting.

Question (Q4: Wigner’s puzzle, as compression). *Why is mathematics unreasonably effective in physics? A conjecture: because both are products of hierarchical compression under scale separation, so the concepts that compress mathematics also compress the world. This is testable. Formalized physics libraries now exist (PhysLean). Does physics reuse the same top concepts as Mathlib, and with the same heavy tail and glue layer?*

7 The growth constant

Naive unfolding grows by a factor of about 1.5 per level:

- 1.45, 1.64 and 1.41 across the three thirds of the height range;
- for comparison, the golden ratio 1.618 for a library citing two predecessors, and 2 for the provable worst case.

Read as a branching process, mathematics is *supercritical*: each level hides about 1.5 times the work of the previous one.

Question (Q5). • *Is this constant a property of mathematics, or of Mathlib’s style?*

- *Is it stable over Mathlib’s history, and the same in Isabelle or Rocq?*
- *Does it relate to the “effective number of essential citations”, via the d-bonacci constants (golden ratio for $d = 2$, approaching 2)?*

8 Delayed reward, measured

The strongest remaining point of the opposing argument: fruitfulness is visible only *decades* later (Lobachevsky), so it cannot be trained on. This is now an experiment.

- Mathlib’s git history records when every declaration was born and when each use appeared.

- If a concept’s reuse after a few months predicts its reuse years later (cumulative advantage), then the effective reward horizon is months, not decades.
- If it does not, the delayed-reward objection gains real empirical support.

Either outcome is informative. I would like to know the answer more than I would like to win the argument.

9 Understanding is conditional compression

Thurston said mathematics is about human understanding. Here is a way to make that precise.

- An agent h *understands* a proof π when π ’s size, *relative to h ’s library of named concepts*, fits within h ’s working capacity.
- This is conditional Kolmogorov complexity, with a library in place of a universal machine.
- It also matches the cognitive science of chunking: chess masters see positions in chunks (Chase and Simon), and memory holds about seven chunks (Miller).

This definition can be plugged into the **Understands** predicate of our earlier paper. It then makes the “fuzzy boundary” concrete. A proof incomprehensible to a human may be comprehensible *given a richer library*. So the question “can humans understand AI mathematics?” becomes “can the AI’s library be taught?”, which is quantitative.

10 Plan (fast, in order of value per hour)

1. **Lean: the sharp d -bonacci bound.** This is conjecture 4 of the README.
 - With at most d citations, $U_i \leq S T_i$, where $T_i = 1 + T_{i-1} + \dots + T_{i-d}$.
 - It is attained by the library citing its d predecessors.
 - Proof idea: the d largest earlier values of a monotone sequence are the last d values. A short exchange argument on finite sets proves it.
2. **Formal paper** with all of the above: theorems, measurements, the correction, and the connections, stated as conjectures where unproved.
3. **Glue and abbreviation analysis**, made rigorous. Classify plumbing by *kind* (instances), not by name, and report the law of abbreviation properly.
4. **Later:** the git-history experiment (Q on delayed reward), and a Stitch-style library-learning rediscovery test on a small part of Mathlib (Q1, “forced vs chosen”).

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